

Smart Prosthetics ERP System Blueprint

Business overview for management, IT, and operations staff · Version 1.0 · August 2026

Client / Center: _____
Document date: _____
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System type: Offline medical ERP
Deployment: Local LAN server
Audience: Non-technical stakeholders

1. Purpose of This Document

This blueprint explains **what the system does**, **who uses it**, and **how a patient case moves** from reception to delivery. It is written for center management, department heads, auditors, and IT staff who need a clear picture without reading source code.

2. System Purpose & Scope

- Smart Prosthetics ERP is an integrated enterprise system for a prosthetics manufacturing center. It manages:
- **Clinical records** — patient registration, appointments, medical exams, measurements
 - **Operational cases** — one case per prosthetic order, tracked through all departments
 - **Inventory** — stock receiving, barcode dispense, bill of materials (BOM)
 - **Finance** — quotes, contract approvals, payments, contract debts, military sovereign debt
 - **Governance** — immutable audit trail and five BI command boards

Offline-first design: The system runs 100% on a local server inside the center. Daily work does **not** require internet. Patient data never leaves the local network. Staff PCs need only a web browser (Chrome recommended).

3. Core Design Principles

Principle	Meaning for the center
Path separation	Civilian and military patients follow different financial rules. Data and reports stay isolated by patient type.
Event-driven workflow	Cases move automatically when a department completes its work. No manual “send to next department” buttons.
QR for people & documents	Patient cards, quotes, and delivery use QR codes for fast, accurate tracking.
Barcode for stock	Every dispense must match the approved BOM. Wrong scans are blocked and logged.
Role-based dashboards	Each employee sees one department screen. Prices and stock balances are hidden where not needed.
Immutable audit log	Every change is recorded permanently. No user can edit or delete audit history.

Patient Pathways

Civilian vs military — two isolated business tracks

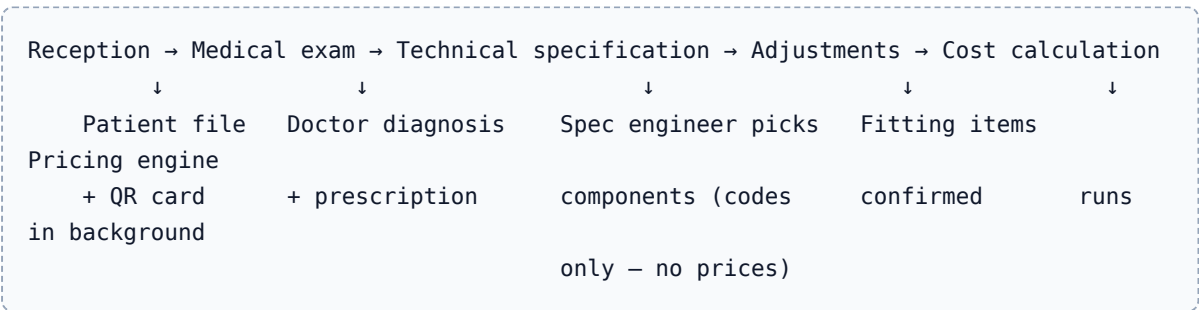
4. Registration & Classification

At reception, every patient receives:

- 1. A permanent **Patient ID** stored in the database
- 2. A printed **patient card with QR code** for queue and status tracking
- 3. A classification: **Civilian** or **Military**

For civilians, the receptionist also links the patient to a **contract company** (insurer, union, employer) with coverage rules. For military patients, rank and sovereign entity are recorded instead.

5. Shared Clinical Path (Both Types)



Blind specification rule

During technical specification, the specialist sees **item codes and quantities only**. Purchase prices and warehouse balances are hidden so clinical decisions stay independent of finance and stock pressure.

6. Civilian Path — Full Financial Gate

Step	What happens
1. Quote issued	PDF quote with amount in words + QR code. Case is financially frozen until approval.
2. Contract approval	Patient obtains official approval letter from contract entity. Reception scans/uploads it (OCR-assisted validation).
3. Operations release	On successful validation, case unfreezes and receives a central work order number .
4. Cash patients (optional)	Patients without a contract company pass through the cashier for payment before manufacturing.
5. Warehouse dispense	Storekeeper scans barcodes against the approved BOM. Mismatch = blocked + audit alert.
6. Manufacturing	Workshop completes production stages (generation → assembly → casting → finishing).
7. Delivery	Reception scans patient QR at delivery. Case closes. Invoice/debt posts to contract entity.

7. Military Path — Financial Bypass

Step	What happens
After cost calculation	No quote and no payment wait . Cost is calculated silently for reporting only.
Services approval (optional)	Certain beneficiary categories may require an internal services approval step before operations.
Operations	Case auto-advances to operations and is auto-approved for manufacturing.
Manufacturing & delivery	Same warehouse barcode rules and workshop stages as civilian path.
Financial close	Cost rolls to sovereign military debt reports — not patient billing.

Important: Civilian and military data must never be mixed in reports, queries, or UI filters without explicit path scoping. This is a non-negotiable security and compliance requirement.

Workflow Stages & Automation

How cases move without manual routing

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8. Main Case Pipeline

```
reception → exam → technical → adjustments → cost_calc  
→ [services_approval] → quote → operations → [cashier]  
→ manufacturing → ready_delivery → delivered → closed
```

Stages in brackets are conditional (pathway configuration or patient type).

9. Event-Driven Transitions

The system does not rely on staff clicking “move to next department.” Instead, completing a domain action triggers the next stage automatically:

Staff action	Automatic result
Doctor approves exam	Case appears in technical specification queue
Spec engineer saves BOM items	Pricing engine runs in background
Costing confirmed (civilian)	Quote generated; case frozen pending approval
Costing confirmed (military)	Case jumps to operations — no quote step
Approval QR scanned (civilian)	Work order released; warehouse can dispense
Barcode dispense completed	Stock updated; workshop notified; WIP started
Workshop marks finished	Case moves to “ready for delivery”
Delivery QR scanned	Case closed; financial posting executed

10. Manufacturing Sub-Stages

Stage	Meaning	BOM status
Warehouse / issue	Materials scanned out of stock	Raw → dispensed
Generation	Initial fabrication	Work in progress (WIP)
Assembly	Component assembly	WIP
Casting	Casting / molding	WIP
Finishing	Final fit, polish, quality check	Finished product
Delivery scan	Patient receives prosthetic	Case financially closed

Note: “Finished product” (BOM complete) and “closed case” (delivered + billed) are two separate steps.

11. SLA Monitoring

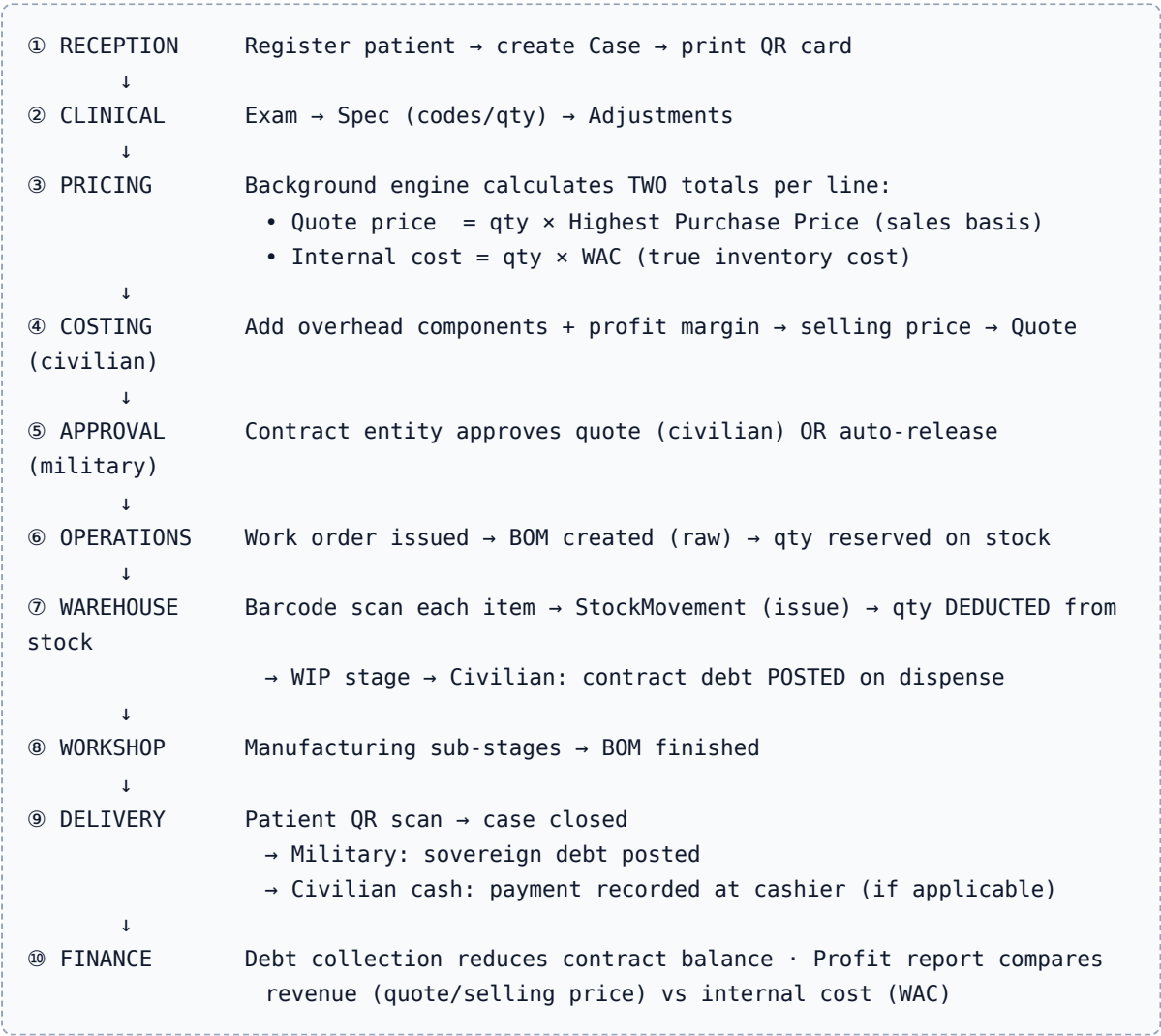
Each case tracks time in stage. Admin BI board #1 flags cases that exceed configured SLA days per department, helping management identify bottlenecks.

Integrated End-to-End Cycle

How clinical, inventory, sales, and finance connect as one chain

Single connected chain: Every department action updates the same case record. Nothing is isolated — stock, sales price, and ledger entries all trace back to one patient case and one work order.

11. Full Business Cycle (Connected)



12. Sales ↔ Stock Link (Critical for Finance)

Event	Stock effect	Financial effect
Goods received (warehouse inward scan)	Qty increases; WAC recalculated; price batch stored	Inventory asset value rises (WAC × qty)
BOM created at operations	Required qty reserved against catalog	No ledger yet — planning only

Barcode dispense (per BOM line)	Qty decremented; <code>StockMovement</code> logged at current WAC ; case <code>issue_cost</code> snapshot stored	Civilian: contract entity debt increased (quote total); cost side = WAC at dispense
Material return (authorized)	Qty restored at WAC unit cost	Reversal / adjustment audited
Delivery closed	No further stock change	Military: sovereign debt entry; civilian: final account closure
Debt collection (finance)	No stock change	ContractCompanyDebt.collected increases; remaining balance drops

13. Multiple Prices Per Item — Why Three Numbers Exist

Price type	Source	Used for
Purchase batches	Each inward receipt creates a <code>StockItemPrice</code> row (supplier, invoice, unit price, qty)	Price history; highest-price lookup
Highest Purchase Price	Max of all batches with qty > 0 (and base catalog price)	Civilian quote / selling price — protects center from repurchase inflation
WAC (Moving Average)	$(\text{prior_qty} \times \text{prior_WAC} + \text{in_qty} \times \text{in_price}) \div (\text{prior_qty} + \text{in_qty})$	Inventory valuation , internal cost, profit margin calculation
Selling price	Materials (highest price) + overhead components + profit %	Patient/entity invoice; revenue in profitability report

Finance clarity: WAC is never mixed into civilian quotes. Quotes use highest purchase price. Profit is measured as *revenue (selling price) minus issue cost (WAC at barcode dispense)*. Admin reports include **Inventory** ↔ **Finance Reconciliation** and **Item Profit Margins** (WAC vs highest price per SKU).

User Roles & Department Dashboards

Who sees what — operational access model

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14. Login & Routing

Each employee has a personal login. After authentication, the system opens **one dashboard** based on role. There is no shared password per department.

15. Role Summary

Role	Primary responsibilities	Financial / stock visibility
Super Admin / Admin	System settings, users, master data, all reports, audit log, pathway config	Full access
Reception	Patient registration, appointments, quote handover, approval scan, delivery QR	Quote totals only — no purchase costs
Doctor	Clinic queue, medical records, exam approval, transfer to spec	Blocked — no prices
Spec engineer	Technical specification — component codes & quantities	Blocked — no prices, no stock qty
Adjustments desk	Fitting / adjustment items before costing	Blocked
Costing desk	Confirm costs, trigger quote (civilian)	Cost view for pricing role
Operations office	Approve/reject cases, release work orders	Operational view
Cashier	Cash payment collection (cash civilians)	Payment amounts
Warehouse (Technical)	Stock receive, BOM barcode dispense, returns	Quantities & specs — no purchase profit view
Workshop	Manufacturing queue, material returns, production stages	Production status

16. Admin — Five BI Command Boards

#	Board	Key indicators
1	Patients & SLA	Total/open cases, civilian/military split, average turnaround, SLA breaches
2	Inventory	Stock value (WAC), low stock alerts, stagnant items (>180 days)
3	Operations	Open work orders, awaiting dispense, in workshop, ready for delivery
4	Entities & costs	Civilian contract debts, military sovereign debt, cash collected — paths separated
5	Purchasing	WAC vs highest purchase price, supplier price trends

17. Tracking Technologies

Technology	Used for	Rule
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QR code	Patient card, quote document, approval scan, delivery close, public status lookup	Human / document workflow
Barcode	Serialized stock movements — receive and dispense	No dispense without validated work order + scan match

Accounting Profile & Financial Reports

For the finance department — inventory ↔ ledger alignment

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18. Four Financial Domains (Period Balances)

The system tracks opening balance, movement, and closing balance for each accounting domain within any date range:

Domain	What it measures	Movement sources
Cash treasury	Cash collected at cashier + debt collections	Payment records, DebtCollectionEntry
Civilian receivables	Amount owed by contract entities	Posted on BOM dispense; reduced on collection
Military sovereign debt	Aggregated military case costs	Posted on delivery; collected separately
Inventory (WAC)	Total stock asset value	Receipts increase value; dispense/returns change qty × WAC

Reports: **Opening Balance** and **Closing Balance** (Admin → Reports). Accounting periods support manual opening overrides for year-start alignment.

19. Profit & Margin Reports

Report	Formula / logic	Audience
Profitability review	Per delivered case: Revenue – issue_cost (WAC at dispense, else internal_cost) = Margin & Margin %	Finance / admin with cost permission
Inventory ↔ Finance reconciliation	Period: receive value, issue value (WAC), delivered revenue, delivered WAC cost, gross margin	Finance — bridges warehouse and P&L
Item profit margins	Per SKU: WAC, highest/lowest purchase price, unit margin, margin %, WAC inventory value	Finance / procurement
Inventory valuation	Each item: qty × WAC and qty × highest price side-by-side	Finance / warehouse
Stock movements	Each movement: qty, unit cost (WAC), line value, BOM reference	Warehouse audit
BI Board 5 — Purchasing	Per item: WAC vs highest price; unit margin % and erosion flag	Procurement

20. Costing Engine → Quote (Civilian Sales Price)

materials_total

+ overhead

= total_cost

= Σ (qty × highest purchase price)

= configurable % components (labour, etc.)

← from pricing engine

```
× (1 + profit_rate%)  
= selling_price    → becomes Quote.total → contract debt basis
```

Military path: same internal WAC cost stored on case; no quote document; sovereign debt uses WAC-based total_cost.

21. Finance Department Controls

- Every stock movement is immutable (`StockMovement`) with WAC unit cost and `balance_after` snapshot
- At dispense, all BOM lines refresh to current WAC — same basis for stock deduction and `issue_cost`
- Every financial posting writes to audit log (tag: financial)
- Duplicate posting prevented (`ledger_posted_at` atomic claim on case)
- Credit notes for post-delivery civilian adjustments (admin approved)
- Civilian and military figures never combined in one report row without explicit label

22. Dual Pricing Policy (Summary)

Purpose	Method	Used when
Inventory valuation & accounting reports	WAC — Weighted Average Cost (moving average)	Stock receipts, inventory reports, profit cost basis
Civilian quotes & case pricing	Highest Purchase Price among batches with qty > 0	Quote generation — protects against inflation / repurchase risk
Profit margin	Selling price – WAC internal cost	Profitability report per delivered case
Military cases	Silent WAC-based background cost	Posted to sovereign debt — never patient invoice

23. Bill of Materials (BOM) Lifecycle

BOM state	Meaning	Stock impact
Raw	Approved material list — awaiting dispense	Qty reserved; not yet deducted
WIP	Materials dispensed to workshop	Qty deducted via barcode; StockMovement issued
Finished	Prosthetic manufacturing complete	Ready for patient delivery

24. Immutable Audit Log

Every create, update, workflow transition, blocked barcode scan, and financial posting is logged automatically with:

- User identity and timestamp (to the second)
- Device identifiers (IP / MAC where available)
- Action type and module tag
- Full before/after data snapshot

Legal-grade trail: Audit records are append-only. No user — including system administrator or IT — can modify or delete entries through the application.

25. Master Data (Configured Once)

- Contract companies and coverage rules
- Military ranks and beneficiary categories
- Stock catalog, categories, suppliers, kits
- Visit types, pathway steps, costing modes
- Users, roles, and permission defaults

26. Document Summary

Smart Prosthetics ERP connects clinical care, manufacturing, inventory, and finance in one offline system. Sales (quotes) deduct from stock only through validated barcode dispense. Multiple purchase prices per item feed both quote pricing (highest) and inventory valuation (WAC moving average). Finance receives opening/closing balances, profitability margins, and purchasing erosion alerts. Civilian and military paths remain fully separated in data and reports.

Related document:

System Architecture (technical deployment & software layers)

Approved by: _____

Signature / date: _____